Assignment 1: NLP Text Evaluation System
U23AI052 Submission

Practical Assignment

Develop an NLP-based Text Evaluation System that accepts a paragraph as input and performs the following operations:

1. Read the input text from the user.
2. Perform **text cleaning and normalization**.
3. Apply **tokenization**.
4. Remove **stop words**.
5. Apply both **stemming and lemmatization**.
6. Display the processed text after each step.
7. Classify the processed text as **Positive or Negative**.
8. Generate a **confusion matrix** using actual and predicted labels.
9. Calculate **Precision, Recall, and F1-Score**.
10. Display the final evaluation results in a clear format.

```
(powermind_rtx5050) D:\Information Retrieval>python tute1.py
C:\Users\HP\miniconda3\envs\powermind_rtx5050\Lib\site-packages\cupy\_environment.py:284: UserWarning: CUD
A path could not be detected. Set CUDA_PATH environment variable if CuPy fails to load.
  warnings.warn(
Enter a paragraph: This is a bad product.

Cleaned text: this is a bad product

Tokens: ['this', 'is', 'a', 'bad', 'product']

After stop word removal: ['bad', 'product']

After lemmatization: ['bad', 'product']

After stemming: ['bad', 'product']

Sentiment: Negative

Confusion Matrix:
[[2 0]
 [0 2]]

Precision: 1.0

Recall: 1.0

F1 Score: 1.0
```